

Shared Doubt: Zero-Shot Cross-Lingual Confidence Estimation for Language Models

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Abstract

Confidence estimation (CE), i.e. quantifying the reliability of a model’s prediction, has attracted great interest in the context of large language models (LLMs). However, most studies focus on English, ignoring the multilingual reality of LLM usage, while many CE methods degrade or require retraining across languages. To address this gap, we investigate whether multilingual LLMs encode shared, language-transferable confidence features in open-ended question answering. We use a lightweight linear probe that predicts answer correctness directly from intermediate representations. Trained monolingually, the probe generalizes zero-shot to *unseen*, typologically diverse languages without target-language supervision. Learned layer weights and multiple ablations reveal that confidence features concentrate in middle layers across languages, suggesting a shared confidence subspace. While zero-shot cross-lingual performance depends on similarity to the source language, the probe provides a strong baseline without any retraining and compares favorably to other popular confidence estimation methods.

1 Introduction

Large language models (LLMs) have seen unprecedented adoption in recent years (Wang et al., 2024b; Humlum and Vestergaard, 2025), including in language communities besides English (Liu and Wang, 2026). Despite both public awareness and academic research on their limitations, *hallucination*, the generation of plausible but factually incorrect statements (see e.g. Huang et al., 2025 for an overview), still poses an ongoing problem that risks downstream harms and the loss of user trust. One potential solution to this problem is confidence estimation (CE), i.e. obtaining numerical values which reflect the probability that the LLM’s output is correct. CE has long been studied in machine learning (DeGroot and Fienberg, 1983; Naeini et al., 2015;

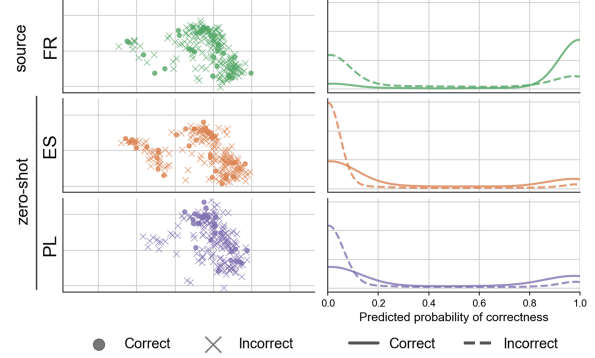


Figure 1: (Left) PCA scatter plots of (in-)correct samples in the probe’s learned aggregation of the LLM latent space. (Right) Distribution of confidence scores.

Guo et al., 2017; Ovadia et al., 2019; Minderer et al., 2021) and has received increasing attention for LLMs (Jiang et al., 2021; Kadavath et al., 2022; Tian et al., 2023; Zhu et al., 2023; Ulmer et al., 2024; Kapoor et al., 2024). However, the vast majority of works focus exclusively on English, leaving other languages largely unexplored. This is particularly concerning as LLMs may behave less reliably in languages with fewer training data, making accurate confidence estimation even more important (Ahuja et al., 2022; Ulmer et al., 2022; Krause et al., 2023).

In this work, we take a first step toward truly multilingual confidence estimation in LLMs, focusing on short-form open-ended question answering (QA). Our approach is motivated by a key observation: prior work on multilinguality has shown that LLMs’ hidden states capture cross-lingual features (Wang et al., 2024a; Wendler et al., 2024; Dumas et al., 2025; Schut et al., 2025), while e.g. Marks and Tegmark (2023a); Ji et al. (2025) have identified uncertainty-related features. We hypothesize that these properties jointly enable cross-lingual transfer of confidence estimates. To test this, we apply a lightweight linear probe that learns a weighted

combination of hidden layers to predict answer correctness. We show that the probe, trained on an LLM’s latent representations in a single language, can predict confidence scores for unseen, typologically diverse languages without any target-language supervision (zero-shot): [Figure 1](#) illustrates how correct and incorrect responses are distributed similarly in the learned aggregation of the target LLM’s latent space. The probe can exploit this fact as shown by its predictive distribution on the right, with some degradation as the zero-shot language becomes more dissimilar.

Our contributions are as follows. To the best of our knowledge, we are the first to demonstrate zero-shot cross-lingual transfer of confidence estimation in open-ended QA through probing of intermediate LLM representations. We analyze how performance varies across unseen languages of the same and different language families and scripts, and show that the probe outperforms baselines from several CE method families *without retraining* on the target language. Through ablations, we find that the most salient information for zero-shot confidence prediction is localized in the middle layers of the model, consistent with prior work suggesting that these layers encode a shared cross-lingual representational space. More broadly, our findings suggest a possible path toward multilingual calibration with little or no target-language supervision, where confidence signals learned in higher-resource languages transfer to lower-resource settings. We release all code as open source.¹

2 Related Work

We position our work at the intersection of studies examining the latent space of multilingual LLMs, the probing of model representations to identify whether specific properties are encoded, and the wider landscape of confidence estimation in LLMs.

Multilingual Latent Spaces. Several studies have examined the organization of LLMs’ latent space, finding that layers closest to the input and output are more language-specific. However, there is no consensus on whether multilingual processing involves mapping to a particular language or through shared, language-agnostic regions in the representation space. Some works support the language dominance hypothesis, according to which inputs are mapped into a representation space

shaped by the most prevalent training language (usually English), perform computations in that space, and translate the result back into the original language ([Zhang et al., 2023](#); [Wendler et al., 2024](#); [Schut et al., 2025](#); [Zhao et al., 2024](#); [Tezuka and Inoue, 2025](#)). [Zhao et al. \(2024\)](#) show that in the middle layers reason in English while leveraging multilingual knowledge, while [Tezuka and Inoue \(2025\)](#) identify two types of neurons, namely ones responsible for mapping from a language-specific space to a shared semantic space in the early and middle layers, and those operating in reverse in the upper layers. [Schut et al. \(2025\)](#) find that semantically meaningful words pass through an English-centric representation space, and [Saji et al. \(2025\)](#) extend these findings to writing systems, suggesting that middle layers represent tokens in Latin script. By contrast, [Zhang et al. \(2024\)](#); [Wang et al. \(2024c\)](#) indicate that multilinguality is facilitated by shared language-agnostic representations, identifying parameters essential to cross-lingual performance. [Bhattacharya and Bojar \(2023\)](#); [Tang et al. \(2024\)](#); [Kojima et al. \(2024\)](#); [Zeng et al. \(2025\)](#) discover language-specific neurons primarily in the lower and upper layers, with a gap in the middle layers. [Dumas et al. \(2025\)](#) further show that the output language is encoded in earlier layers, whereas the concept to be translated emerges in later layers. [Shani and Basirat \(2025\)](#) find no consistently dominant language in the middle layers.

Probing Latent Representations. Probes are lightweight models that map a model’s internal representations to a property of interest ([Alain and Bengio, 2017](#); [Belinkov, 2022](#)), and are widely used in mechanistic interpretability to test whether a property is encoded in LLM representations ([Bereska and Gavves, 2024](#); [Ferrando et al., 2024](#)). In multilingual LLM research, probes have been used to study morphosyntax ([Brinkmann et al., 2025](#); [Wang et al., 2024a](#)), factual knowledge ([Li et al., 2024](#)), semantics ([Li et al., 2024](#); [Wang et al., 2024a](#)), and context ([Ju et al., 2024](#)), with shared features typically emerging in the middle layers. Relevant to our work, probes have been trained on LLM representations in English to detect truthfulness ([Li et al., 2023](#); [Marks and Tegmark, 2023a](#); [He et al., 2024](#); [Azaria and Mitchell, 2023a](#)), with [Li et al. \(2024\)](#) finding stronger signals in high-resource than in low-resource languages. While [Li et al. \(2023\)](#); [Marks and Tegmark \(2023a\)](#) find that truthfulness signals peak in the middle layers, [Jin](#)

¹The code repository can be found under github.com/AthinaKyriakou/shared-doubt.

et al. (2025) show that more complex properties require deeper layers.

Confidence Estimation in LLMs. As mentioned in the introduction, confidence estimation (CE), i.e. obtaining a score that reflects the probability that a model’s answer is correct, has recently attracted a flurry of new work (see e.g. Baan et al., 2023; Huang et al., 2024; Ulmer, 2024; Liu et al., 2025; Geng et al., 2024; Shorinwa et al., 2025). Since some methods rely on the generation process itself, for example token logits or additional model responses, they are immediately applicable to any language, albeit without any guarantee of performance. Other methods, by contrast, are trained on a specific language. Applying them to unseen languages therefore constitutes a zero-shot prediction problem. Probing approaches, which have also been applied to CE (Kadavath et al., 2022; Marks and Tegmark, 2023a; Burns et al., 2023; Liu et al., 2024a; Xiong et al., 2024b; Liu et al., 2024c; Fathullah et al., 2024; Shelmanov et al., 2025), fall into this setting. Importantly, however, few works test performance beyond English, especially on lower-resourced languages. Krause et al. (2023) show that the calibration of verbalized confidence scores varies greatly across languages, with Yang et al. (2023) showing the same for sequence-likelihood-based confidence. Xue et al. (2025) introduce a multilingual CE benchmark and also report wide performance differences across existing methods. In all three, however, confidence is read from model outputs rather than internal representations, and any improvement relies on target-language data or language-specific prompting. None demonstrates zero-shot prediction. Closer to our approach, Zhou et al. (2025) find that the most salient features for multilingual CE lie in the middle layers, but restrict themselves to multiple-choice question answering and do not consider zero-shot prediction.

3 Methodology

Based on the insights from the literature, we now discuss which representations to consider, verify our assumptions, and define the probe.

3.1 Selected Intermediate Representations

Our work focuses on decoder-only transformer-based LLMs (Vaswani et al., 2017; Radford et al., 2019). Given an input token sequence, the model embeds the token at every time step t into a high-

dimensional space $\mathbf{x}_t \in \mathbb{R}^D$ and produces a hidden state $\mathbf{h}_t^{(\ell)}$ by running sequentially through each block of transformer block at layer ℓ , before finally generating sequence $y_1, y_2, \dots, y_T$. Each block updates the previous hidden states in a residual way:

$$\mathbf{h}_t^{(\ell)} = \mathbf{h}_t^{(\ell-1)} + g_\ell(\mathbf{h}_1^{(\ell-1)}, \dots, \mathbf{h}_{t-1}^{(\ell-1)}), \quad (1)$$

where g_ℓ is the model-specific transformation of the ℓ -th block applied to the prefix and current token. This is also referred to as the residual stream, serving as a central pathway that carries and incrementally refines each token’s representation.

Following Zhou et al. (2025), we exploit signals from all layers but focus on the hidden states at two time steps: the prompt’s last token (*last query token*) and the generated answer’s last token (*last answer token*). The former serves as a proxy for the entire input, integrating information from all preceding tokens and capturing the model’s confidence at the start of generation. This token is widely used in multilingual interpretability and uncertainty studies (Wendler et al., 2024; Dumas et al., 2025; Wang et al., 2024c; Brinkmann et al., 2025; Li et al., 2024; He et al., 2024; Duan et al., 2024; Ji et al., 2024). The latter represents the model’s state after generation and may encode its confidence in the completed response. In other contexts it has been shown to encode knowledge (Burns et al., 2022) or truthfulness (Azaria and Mitchell, 2023b).

3.2 A Cross-lingual Confidence Probe

Prior work has identified considerable overlap of latent representations for different language representations in the middle layers (Wendler et al., 2024; Wang et al., 2024a; Zhou et al., 2025; Zhang et al., 2024). We confirm this in our case, showing a selection of PCA plots in Figure 2. Since that implies that some layers will carry more generalizable information, we define a probe that learns a weighted average of layer hidden representations $\mathbf{h}_t^{(\ell)}$ by introducing a learnable parameter $\mathbf{w} \in \mathbb{R}^L$, using it to compute layer weights α_ℓ and pooled representation $\bar{\mathbf{h}}_t \in \mathbb{R}^D$:

$$\alpha_\ell = \text{softmax}(\mathbf{w})_\ell = \frac{\exp(w_\ell)}{\sum_{k=1}^L \exp(w_k)}, \quad (2)$$

$$\bar{\mathbf{h}}_t = \sum_{\ell=1}^L \alpha_\ell \mathbf{h}_t^{(\ell)}. \quad (3)$$

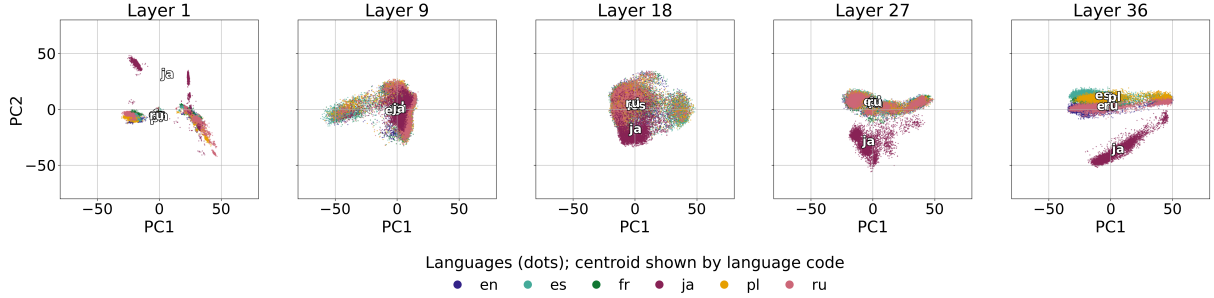


Figure 2: Organization of latent representations in Qwen 3 8B on Global-MMLU, using the last query token.

This representation allows the probe to select the most informative layers for confidence estimation during training. It further assumes that each dimension is approximately consistent across layers, a side effect of Equation (1) as information propagates through layers. Finally, a linear layer with learnable parameters $\mathbf{v} \in \mathbb{R}^D, b \in \mathbb{R}$ and the sigmoid function $\sigma(\cdot)$ are applied to obtain a confidence score $\hat{p}_t$:

$$\hat{p}_t = \sigma(\mathbf{v}^\top \bar{\mathbf{h}}_t + b). \quad (4)$$

While non-linear designs of the probe are possible, evidence from the linear representation hypothesis (Bereska and Gavves, 2024) and other studies (Park et al., 2023; Jiang et al., 2024; Ji et al., 2025; Merullo et al., 2025), suggests that features are often represented linearly in latent space.

4 Experiments

4.1 Setup

Datasets. We base our experiments on two popular parallel question-answering datasets that enable comparable evaluation across languages. The first is MKQA (Longpre et al., 2021), which provides open-ended questions across 26 languages, with the original English questions extracted from Natural Questions (Kwiatkowski et al., 2019). We consider short-phrase, entity, numeric, numeric-with-unit, and date question types and exclude time-sensitive questions, retaining at most 5237 per language. Further details of the selection process are in Section B.1.1. The second is Global-MMLU (Singh et al., 2024), a multilingual multiple-choice dataset covering 42 languages, derived from the English MMLU dataset (Hendrycks et al., 2021). Since multiple-choice evaluation is discriminative and not representative of real-world generative scenarios (Chandak et al., 2025), we keep only questions that can be posed in an open-ended manner, fol-

Language	Genus	Script	MKQA translation	Global-MMLU translation
English	Germanic	Latin	N/A	N/A
French	Romance	Latin	Human	Human
Spanish	Romance	Latin	Human	Human
Polish	Slavic	Latin	Human	Google Translate
Russian	Slavic	Cyrillic	Human	Human & Google Translate
Japanese	Japanese	Japanese	Human	Human

Table 1: Languages and translation types by dataset.

lowing the process detailed in Section B.1.2. This yields a final set of 4215 questions per language.

Selected Languages. To capture cross-lingual variation, we select the six languages shown in Table 1: English (en) is included as it dominates most LLMs’ training data. We further include French (fr) and Spanish (es), two closely related languages that belong to the Romance genus (Dryer and Haspelmath, 2013) and use the Latin script. Polish (pl) and Russian (ru) are chosen to represent the more dissimilar Slavic languages (Dryer and Haspelmath, 2013), with Russian additionally introducing the Cyrillic script. Finally, Japanese (ja) differs in both linguistic family and script. We choose French rather than English as the source language for training the probe: this lets us test on Spanish as a near-out-of-distribution case and on English as a more typologically distant, training-dominant one, alongside the other languages we consider.²

Models. We test our approach on two widely used multilingual auto-regressive LLMs, Llama 3.1 8B Instruct (Grattafiori et al., 2024) (32 layers) and Qwen3 8B (Qwen Team, 2025) (36 layers), the latter run in non-thinking mode for direct short-form answers. We generate with the recommended temperatures (0.6 for Llama, 0.7 for Qwen) and a maximum of 40 new tokens. The per-dataset

²Our main results with English as the source language are included in Tables 6 and 7 in Section A.1.

Dataset	LLM	fr	en	es	pl	ru	ja
MKQA	Llama 3.1 8B	.27	.38	.24	.26	.17	.14
	Qwen3 8B	.18	.26	.18	.16	.13	.12
Global-MMLU	Llama 3.1 8B	.24	.35	.24	.19	.19	.17
	Qwen3 8B	.27	.40	.26	.23	.22	.22

Table 2: LLM accuracy per dataset and language.

and per-language system prompts are given in Figure 16 in Section B.2. All experiments were run on a single NVIDIA A100-80G-PCIe GPU.

Answer Correctness. We assess answer correctness with GPT-4.1 models as judges (OpenAI, 2025), prompted as shown in Figure 17. Further details, including the judge model used per dataset and language, are in Section B.2. We obtain the accuracies in Table 2, with two patterns being notable: accuracies differ across languages, reflecting differences in factual consistency (Fierro and Søgaard, 2022; Qi et al., 2023), and performance is strongest in English and degrades roughly with typological distance from it.

Evaluation Metrics. We evaluate confidence estimates with standard metrics. For discrimination, i.e. how well they separate correct from incorrect responses, we report the area under the receiver operating characteristic (AUROC) and the precision-recall curve (AUPR). For calibration, i.e. how well confidence scores match the probability of correctness, we report expected calibration error with 10 bins (ECE; Naeini et al., 2015; Guo et al., 2017) and the Brier score (Glenn et al., 1950).

4.2 Zero-Shot Confidence Estimation Results

We first evaluate the probe’s zero-shot CE performance in isolation.³ We include a majority-class baseline (*majority*), which in our case is the negative class, and a *prior-probability* baseline, which draws binary correctness labels from a Bernoulli distribution with parameter p set to the training accuracy. We also train a probe only on examples from the target language (*oracle* probe).

Results. We show detailed results for Qwen 3 8B in Table 3 and visualize the overall metric trends in Figure 3, with complementary results for Llama 3.1 8B in Section A.1. We observe the following. First, the *majority* and *prior probability* baselines can appear well-calibrated on ECE, which is not a proper

scoring rule (Liu et al., 2023a), yet they fail to discriminate correct from incorrect predictions⁴. Second, the probe performs well when tested on its training language, French. Third, the degradation from French to more distant languages is clearer for discrimination (AUROC, AUPR) than for calibration (Brier, ECE). Discrimination falls in a fairly consistent, roughly distance-related manner, as exemplified by the difference to the oracle, whereas calibration changes are less systematic, in some settings remaining stable and in others degrading sharply. Nonetheless, in many settings the zero-shot probe stays close to the target-language oracle, and occasionally matches or exceeds it, indicating that cross-lingual transfer of correctness features is often viable despite these drops. The more systematic, distance-related decline in discrimination without a corresponding calibration trend is especially visible in Figure 3. This asymmetry is consistent with zero-shot transfer being affected less by covariate shift in the multilingual latent space than by concept shift in the conditional label distribution (Moreno-Torres et al., 2012), which may reflect the lower target LLM accuracy on languages such as Russian and Japanese. Lastly, the last answer hidden states yield better discrimination and the last query token lower calibration error. The former is expected, as the query latents do not encode information about the generated answer.

4.3 Comparison To Other Estimators

We further compare our probe (Cross-ling. Probe) to other common CE methods.⁵ We consider the length-normalized sequence likelihood (Seq. Likelihood), P(True) (Kadavath et al., 2022), which asks whether a response was true or false and considers the probability of the “true” token, verbalized confidence (Verbalized Conf.; Lin et al., 2022; Xiong et al., 2024a; Ulmer et al., 2025), which elicits a confidence score from 0-100, prompted in the target language, and the mass-mean probe by Marks and Tegmark (2023b), which is not trained but directly derived from the centroids of correct and incorrect predictions.⁶ Prompts for P(True) and

⁴We compute AUPR by trapezoidal integration (`sklearn.metrics.auc`), which inflates it for constant-prediction baselines. Average precision (equal to the positive-class prevalence) is a more appropriate reference, and closely matches trapezoidal AUPR for non-degenerate methods (Boyd et al., 2013).

⁵We only show results for the last answer token for comparability. See Section A.2 for the last query token results.

⁶Compared to Marks and Tegmark, we do not use patching, but apply their probe directly to the middle-layer latents of

³Details on probe training are given in Section B.4.

		MKQA				Global-MMLU			
		AUROC↑	AUPR↑	Brier↓	ECE↓	AUROC↑	AUPR↑	Brier↓	ECE↓
all*	Majority	.50±.00	.58±.02	.17±.05	.17±.05	.50±.00	.63±.03	.26±.06	.26±.06
	Prior prob.	.50±.00	.58±.02	.14±.03	.04±.03	.50±.00	.63±.03	.19±.03	.06±.03
fr	Last Query	.73±.01	.36±.01	.17±.01	.14±.00	.71±.01	.49±.02	.21±.01	.16±.01
	Last Answer	.82±.01	.49±.02	.14±.01	.12±.01	.78±.00	.58±.01	.19±.00	.16±.00
en	Last Query	.70±.01 +.00	.47±.01 +.03	.22±.01 -.01	.18±.01 +.00	.64±.02 -.07	.51±.02 -.09	.31±.01 +.06	.26±.01 +.07
	Last Answer	.70±.01 -.08	.44±.01 -.13	.24±.01 +.06	.22±.01 +.09	.72±.01 -.04	.60±.01 -.09	.27±.01 +.04	.24±.01 +.05
es	Last Query	.71±.01 -.01	.34±.01 -.02	.17±.00 -.01	.14±.01 -.02	.68±.01 +.00	.41±.02 -.03	.27±.02 +.05	.25±.03 +.07
	Last Answer	.82±.01 -.01	.53±.03 +.01	.19±.02 +.05	.20±.03 +.09	.74±.01 -.03	.52±.01 -.03	.23±.01 +.03	.21±.02 +.04
pl	Last Query	.67±.01 -.06	.28±.01 -.06	.15±.01 -.01	.13±.01 -.01	.73±.01 +.01	.40±.01 -.02	.20±.01 -.01	.17±.02 +.00
	Last Answer	.73±.01 -.09	.35±.03 -.13	.23±.03 +.10	.24±.03 +.13	.75±.01 -.04	.48±.01 -.04	.27±.03 +.09	.27±.03 +.11
ru	Last Query	.72±.01 +.03	.30±.01 +.02	.12±.00 -.04	.11±.01 -.05	.68±.01 -.05	.35±.02 -.11	.23±.02 +.04	.21±.03 +.05
	Last Answer	.80±.01 +.01	.43±.03 +.00	.15±.03 +.03	.16±.04 +.05	.72±.01 -.08	.43±.01 -.13	.39±.03 +.22	.42±.03 +.28
ja	Last Query	.57±.02 -.21	.16±.01 -.17	.13±.01 +.00	.11±.01 +.00	.64±.01 -.10	.32±.02 -.13	.20±.02 +.01	.17±.03 +.00
	Last Answer	.64±.04 -.23	.23±.04 -.30	.16±.02 +.07	.16±.02 +.09	.72±.01 -.07	.44±.02 -.08	.31±.02 +.13	.31±.02 +.15

Table 3: Results for Qwen 3 8B on MKQA and Global-MMLU. Top block: baselines (*majority*, *prior prob.*), reported as the average across all languages (*all**), and the probe trained and tested on French (*fr*). Lower block: zero-shot performance of the French-trained probe on each test language. *Last Query* / *Last Answer* = probe trained on last query / last answer hidden states. Values are mean $\pm$ std. across seeds. Gray numbers give the difference to the target-language oracle probe, i.e. $\Delta < 0$ indicates that the oracle scored higher.

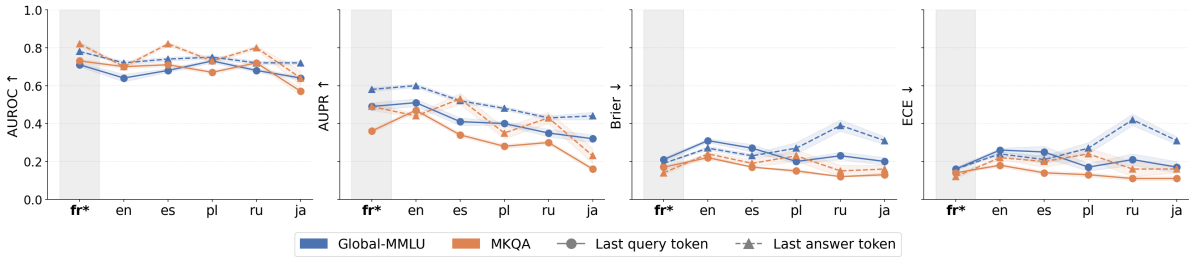


Figure 3: Cross-lingual generalization of the probe trained on French (*fr**) for Qwen 3 8B. Each panel compares the last query token (circle, solid) against the last answer token (triangle, dashed) across test languages, for Global-MMLU (blue) and MKQA (orange). Shaded bands show ± 1 standard deviation.

verbalized confidence are shown in Section B.2. Note that, apart from the mass-mean probe, the baselines are not zero-shot in the sense of having an explicit train-test distribution shift. They are yet intuitive choices for CE in unseen languages.

Results. Table 4 reports results of the last token answer hidden states for selected languages, with the full evaluation in Section A.2. At the closest target, Spanish, the probe is the strongest estimator overall, leading on AUROC and remaining competitive on AUPR and calibration. On the more distant Slavic languages, it remains competitive on both discrimination and calibration in most setups. Discrimination is weakest for Japanese, the correct and incorrect generations.

most distant target, where baselines outperform the probe on MKQA but none dominates across setups. Relative to the baselines, calibration is the probe’s most consistent dimension. Nonetheless, the probe pairs competitive discrimination and calibration. Sequence likelihood discriminates comparably but is miscalibrated, verbalized confidence discriminates only moderately and is the most poorly calibrated of all methods, the mass-mean probe discriminates competitively in places but is inconsistently calibrated, and $P(\text{True})$ is well-calibrated but ranks poorly. That both the cross-lingual and mass-mean probes recover transferable signal supports the hypothesis of language-agnostic confidence features, and suggests that calibration

		MKQA				Global-MMLU			
		AUROC↑	AUPR↑	Brier↓	ECE↓	AUROC↑	AUPR↑	Brier↓	ECE↓
es	Seq. Likelihood	.81±.00	.52±.00	.51±.00	.61±.00	.70±.00	.51±.00	.53±.00	.59±.00
	P(True)	.30±.00	.13±.00	.19±.00	.19±.00	.41±.00	.23±.00	.26±.00	.26±.00
	Mass-Mean Probe	.65±.00	.52±.00	.45±.00	.45±.00	.59±.00	.50±.00	.41±.00	.41±.00
	Verbalized Conf.	.75±.00	.51±.00	.54±.00	.59±.00	.73±.00	.55±.00	.52±.00	.58±.00
	Cross-ling. Probe	.82±.01	.53±.03	.19±.02	.20±.03	.74±.01	.52±.01	.23±.01	.21±.02
ru	Seq. Likelihood	.79±.00	.35±.00	.49±.00	.62±.00	.75±.00	.47±.00	.54±.00	.62±.00
	P(True)	.43±.00	.12±.00	.12±.00	.12±.00	.45±.00	.21±.00	.21±.00	.21±.00
	Mass-Mean Probe	.73±.00	.51±.00	.40±.00	.41±.00	.62±.00	.57±.00	.51±.00	.51±.00
	Verbalized Conf.	.72±.00	.36±.00	.55±.00	.61±.00	.73±.00	.49±.00	.51±.00	.57±.00
	Cross-ling. Probe	.80±.01	.43±.03	.15±.03	.16±.04	.72±.01	.43±.01	.39±.03	.42±.03
ja	Seq. Likelihood	.81±.00	.41±.00	.43±.00	.56±.00	.74±.00	.49±.00	.48±.00	.57±.00
	P(True)	.60±.00	.16±.00	.13±.00	.13±.00	.54±.00	.24±.00	.21±.00	.21±.00
	Mass-Mean Probe	.83±.00	.57±.00	.22±.00	.23±.00	.52±.00	.59±.00	.73±.00	.73±.00
	Verbalized Conf.	.67±.00	.37±.00	.66±.00	.72±.00	.72±.00	.44±.00	.57±.00	.63±.00
	Cross-ling. Probe	.64±.04	.23±.04	.16±.02	.16±.02	.72±.01	.44±.02	.31±.02	.31±.02

Table 4: Comparison of zero-shot confidence estimation methods for Qwen3 8B on MKQA and Global-MMLU on selected languages.

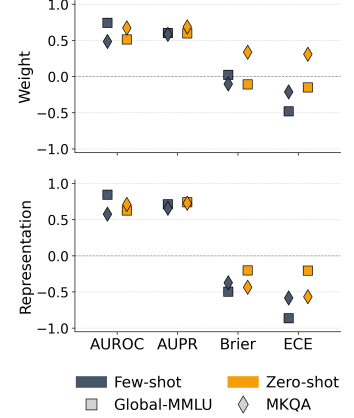


Figure 4: Spearman’s ρ between layer importance and per-layer ablation impact for last token answer for Qwen 3 8B. (Top): weight-based; (Bottom): representation-based.

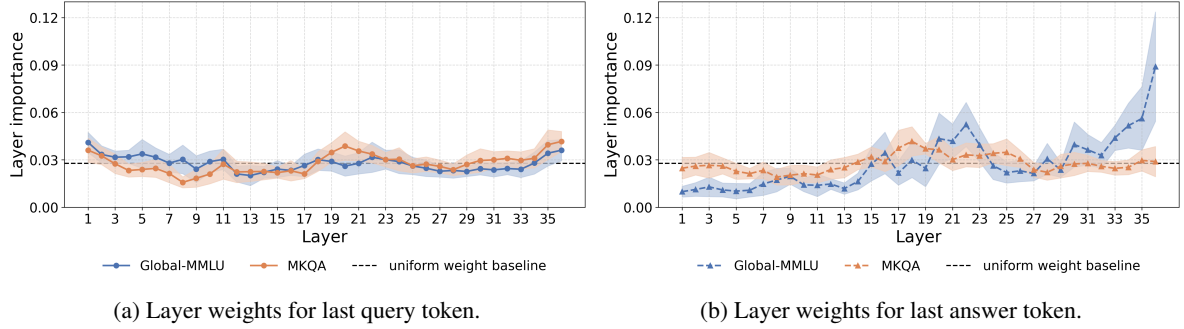


Figure 5: Distribution of learned layer weights from probes trained with different random seeds on the hidden states of the last (a) query token and (b) answer token for Qwen 3 8B on Global-MMLU and MKQA. Shaded areas indicate ± 1 standard deviation, dashed line shows uniform weight.

may benefit from improved probe training data and architecture or from post-hoc recalibration methods such as temperature scaling (Guo et al., 2017) or isotonic regression (Oron and Flounoy, 2017).

4.4 Understanding Layer Importance

To understand what makes the probe work, we first examine the learned layer weights in Figure 5 (Llama 3.1 8B in Section A.3). The weight distribution is more uniform for the last query token, whereas for the last answer token both datasets down-weight early layers in favor of middle ones. To test whether these weights *causally* affect performance, we apply the interventional framework of Geiger et al. (2025), assessing the importance of layer ℓ via the performance gap on a metric $\mathcal{P}$ between a model $\mathcal{M}$ and a variant $\mathcal{M}^{\setminus \ell}$ in which

only the component ℓ is *changed*:

$$\Delta(\mathcal{M}, \ell) := \mathcal{P}(\mathcal{M}) - \mathcal{P}(\mathcal{M}^{\setminus \ell}), \quad (5)$$

where positive values of Δ indicate that the component ℓ contributes positively to model performance. We apply this framework in two ways: by setting specific weights $\alpha_\ell = 0$ and re-normalizing the distribution (*weight ablation*), and by shuffling each layer’s representations across examples (*representation ablation*).⁷ Further details are provided in Section A.4.

Results. We show a selection of results in Figure 6, with the rest given in Section A.4. We observe that removing middle layers leads to a less performant probe, with only the ablation of later

⁷Given the redundancy of adjacent transformer layers (Men et al., 2025), we use a sliding window of size 3.

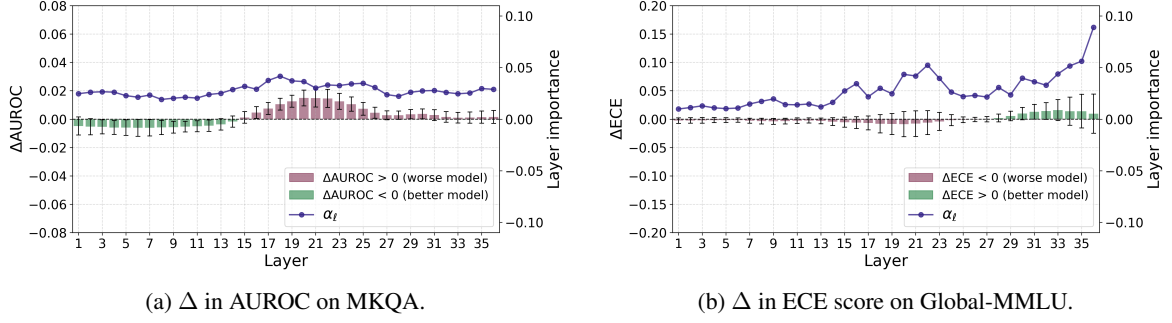


Figure 6: Impact of layer ablation on zero-shot AUROC and ECE (avg. over all languages) for Qwen3 8B. Bars show mean change in metric value over 10 random seeds. Whiskers denote ± 1 standard deviation.

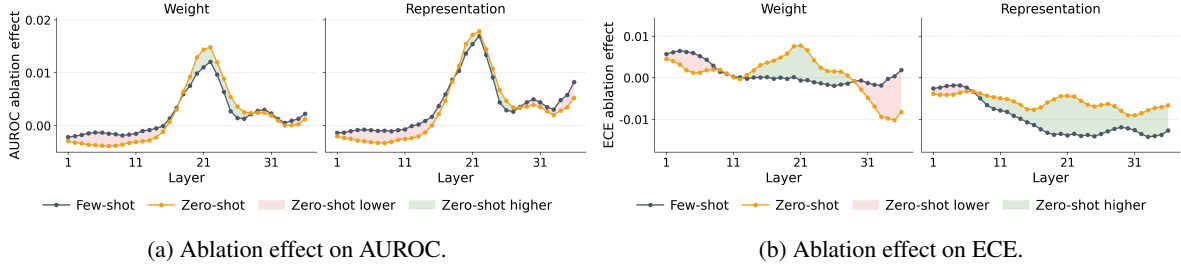


Figure 7: Ablation effect of layers on (a) AUROC and (b) ECE by few-shot (fr) and zero-shot results for Qwen 3 8B and probe trained on last answer token hidden states.

layers yielding improvements in some cases, likely due to the probe learning language-specific information about the source that is not useful in a zero-shot setting. Measuring Spearman’s ρ between layer weights and ablation impact per metric in Figure 4, we find a correlation between layer weights and ablation impact that is preserved in zero-shot settings, though the effect is less consistent on calibration. Plotting the layer ablation impact separately for the few-shot vs. zero-shot setting in Figure 7, we find that the middle-layer ablation has a stronger degrading effect on calibration in the zero-shot setting, while its effect on discrimination remains similar across both settings. Lastly, Section A.5 shows that while learnable layer weights help localize language-agnostic signals, the probe recovers much of the signal directly from the latents, likely due to the layer redundancy visualized in Figure 2. While the comparison between the uniform-weight and learned-weight probes is inconsistent, we view this, together with Zhou et al. (2025), as evidence that more robust future approaches will likely rely on specific layers.

5 Discussion

Our results indicate that the language-general spaces in multilingual LLMs found by prior work (e.g. Wendler et al., 2024; Zhao et al., 2024; Schut

et al., 2025) carry useful signal for confidence estimation. While Zhou et al. (2025) show that the best representations for multilingual confidence estimation indeed lie in middle layers, something that our ablations in Section 4.4 confirm, we are the first to demonstrate that this information can be used *zero-shot* on unseen languages. The probe performs strongly when the target language is close to the source. As similarity to the source decreases (Section 4.2), discrimination degrades roughly with linguistic distance, while calibration changes less systematically, being stable in some settings but degrading in others. Nevertheless, among common CE methods in Section 4.3, the probe combines competitive discrimination and calibration in many setups. We also found that the choice of time step matters. The last query token, often used in related probing work (e.g. Wendler et al., 2024; Dumas et al., 2025; Wang et al., 2024c; Brinkmann et al., 2025), was better calibrated on unseen languages, but on the source language the lesser-used last answer token (Burns et al., 2022; Azaria and Mitchell, 2023a) achieved both better calibration and discrimination. This is possibly because query latents encode question difficulty, while answer latents reflect the response actually given.

6 Conclusion

In this work, we presented the first evidence that language-agnostic latent spaces in multilingual LLMs contain correctness signals that can be exploited for confidence estimation on unseen languages, without any target-language supervision. Consistent with prior work, we find this signal concentrated in the middle, language-general layers. While transfer weakens as the target language grows more distant from the source, the probe remains a strong and practical baseline across typologically diverse languages, and, among the estimators we compare, most consistently pairs competitive discrimination with calibration. It remains especially interesting to explore how to make this approach more robust to unseen language families and scripts by training on multiple source languages, identifying the most critical latent representations, and applying common post-hoc calibration techniques. More broadly, we believe that these findings open research directions towards multilingual confidence estimation with little or no supervision, especially for underserved languages.

Limitations

We acknowledge the following limitations. First, our work focuses on open-source models of 8 billion parameters. We nonetheless expect our approach to extend to larger ones. The localization of language-agnostic (Zeng et al., 2025; Schut et al., 2025; Wendler et al., 2024; Tang et al., 2024; Liu et al., 2024b) and correctness-related (Khanmohammadi et al., 2025; Bao et al., 2025; Marks and Tegmark, 2023a) signals follows consistent patterns across model scales. We do not, however, test this directly, and transfer behavior at scale remains an open question. Second, while our language selection spans different families and scripts, it remains weighted toward European, relatively high-resource languages. We expect zero-shot performance to degrade further for languages that are less related to the source, use a different script, or are less represented in training, consistent with the greater representational dissimilarity observed for more distant languages (Li et al., 2024; Kudugunta et al., 2019; Singh et al., 2019). Third, correctness labels are generated using an LLM-as-a-judge, whose reliability may vary across languages, potentially introducing language-correlated label noise. Fourth, we focus on short-form QA, whereas recent works have also started to consider reason-

ing (Zhang et al., 2025c; Yoon et al., 2025; Zhang et al., 2025a; Razghandi et al., 2025; Mao et al., 2026) and long-form uncertainty quantification (Liu et al., 2023b; Zhang et al., 2025b; Fan et al., 2026; Bouchard et al., 2026). Our approach may extend to these settings, but longer generations likely require more careful selection of the hidden-state time step. Finally, affine probes may be insufficient in some cases. Self-attention and feed-forward blocks can substantially rotate and mix representation dimensions, so readouts may miss non-linearly encoded structure, while our weighted average of layer representations may not capture relevant cross-layer interactions.

Ethical Considerations

We foresee no immediate ethical issues.

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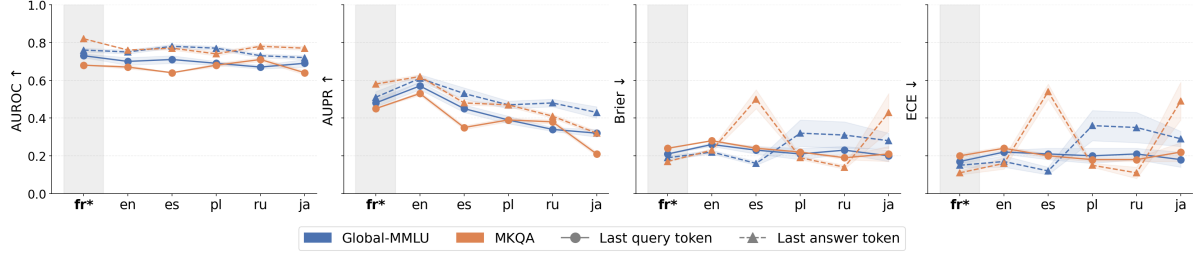


Figure 8: Cross-lingual generalization of the probe trained on French (fr*) for Llama 3.1 8B. Each panel compares the last query token (circle, solid) against the last answer token (triangle, dashed) across test languages, for Global-MMLU (blue) and MKQA (orange). Shaded bands show ± 1 standard deviation.

		MKQA				MMLU			
		AUROC↑	AUPR↑	Brier↓	ECE↓	AUROC↑	AUPR↑	Brier↓	ECE↓
all*	Majority	.50±.00	.62±.04	.24±.08	.24±.08	.50±.00	.62±.03	.23±.06	.23±.06
	Prior prob.	.50±.00	.62±.04	.18±.04	.07±.06	.50±.00	.62±.03	.18±.03	.05±.04
fr	Last Query	.68±.00	.45±.00	.24±.00	.20±.01	.73±.01	.48±.02	.21±.00	.17±.01
	Last Answer	.82±.00	.58±.01	.17±.00	.11±.01	.76±.01	.51±.03	.19±.01	.15±.01
en	Last Query	.67±.01 +.02	.53±.01 +.00	.28±.00 +.00	.24±.01 +.02	.70±.02 +.01	.57±.02 +.05	.26±.01 +.00	.22±.01 +.00
	Last Answer	.76±.00 -.02	.62±.01 -.05	.23±.01 +.03	.16±.03 +.03	.75±.01 -.02	.61±.02 -.05	.22±.01 +.01	.17±.03 +.01
es	Last Query	.64±.00 +.04	.35±.01 +.00	.24±.01 -.03	.20±.01 -.04	.71±.02 +.05	.45±.02 +.07	.23±.02 +.00	.21±.03 +.01
	Last Answer	.77±.01 -.04	.48±.01 -.10	.50±.05 +.33	.54±.04 +.42	.78±.01 +.00	.53±.03 -.04	.16±.01 -.01	.12±.02 -.01
pl	Last Query	.68±.01 +.00	.39±.01 -.03	.22±.00 -.01	.18±.01 -.01	.69±.01 -.01	.39±.02 +.03	.21±.03 +.03	.20±.04 +.04
	Last Answer	.74±.01 -.05	.47±.01 -.09	.19±.01 +.01	.15±.01 +.02	.77±.01 +.01	.47±.02 +.01	.32±.07 +.16	.36±.08 +.22
ru	Last Query	.71±.01 +.05	.38±.01 +.05	.19±.01 +.00	.18±.01 +.00	.67±.01 -.09	.34±.01 -.11	.23±.02 +.06	.21±.03 +.07
	Last Answer	.78±.01 -.05	.41±.01 -.13	.14±.01 +.00	.11±.03 +.00	.73±.01 -.06	.48±.02 -.05	.31±.07 +.15	.35±.08 +.22
ja	Last Query	.64±.01 -.09	.21±.01 -.08	.21±.04 +.05	.22±.05 +.08	.69±.02 +.05	.32±.02 +.06	.20±.03 +.01	.18±.04 +.01
	Last Answer	.77±.01 -.03	.32±.01 -.10	.43±.10 +.30	.49±.10 +.39	.72±.01 -.01	.43±.03 +.02	.28±.04 +.11	.29±.04 +.13

Table 5: Results for Llama 3.1 8B on MKQA and Global-MMLU. First group shows the results for the probe and baselines trained and tested on French. all* = avg. across all languages. Remaining groups list zero-shot performance on test languages. Last Query / Last Answer = Probe trained on last query / answer hidden. Gray numbers state performance difference compared to the oracle probe, i.e. $\Delta < 0$ indicating that the oracle scored higher.

Findings of the Association for Computational Linguistics: EMNLP 2023, pages 9778–9795.

A Additional Results

A.1 Confidence Estimation Results

This section contains complete results for the confidence estimation in Section 4.2. Table 5 shows the results with French as the source language for Llama 3.1. Tables 6 and 7 gives both sets of results, but with English used as the source language.

A.2 Comparison to Other Estimators

This section contains the full results for the comparison with other confidence estimators in Section 4.3. This includes first the full results for all languages, including the few-shot setting in French, in Tables 8

and 9 for Qwen 3 and Llama 3.1 respectively. In Section 4.3 we chose to compare other estimators to our probe trained on last answer token hidden representations, since other methods also, implicitly or explicitly, have access to full generation of the model. To also benchmark the probe trained on the last input token hidden representations, we provide Tables 8 and 9.

A.3 Additional Layer Weight Plots

The corresponding layer weight plots for Figure 5 in Section 4.4 for Llama 3.1 are given in Figure 9.

A.4 Additional Ablation Results

This section details the ablation methodology of Section 4.4 and provides additional results.

		MKQA				MMLU			
		AUROC↑	AUPR↑	Brier↓	ECE↓	AUROC↑	AUPR↑	Brier↓	ECE↓
en	Last Query	.65±.01	.53±.01	.28±.00	.22±.01	.69±.01	.52±.01	.26±.01	.22±.01
	Last Answer	.78±.01	.67±.01	.20±.00	.13±.01	.77±.01	.66±.01	.21±.01	.16±.02
fr	Last Query	.63±.01 -.05	.34±.01 -.11	.27±.01 +.03	.22±.00 +.02	.66±.01 -.07	.35±.02 -.13	.23±.02 +.02	.21±.03 +.04
	Last Answer	.76±.01 -.06	.56±.01 -.02	.24±.04 +.07	.23±.06 +.12	.74±.02 -.02	.46±.03 -.05	.22±.03 +.03	.20±.05 +.05
es	Last Query	.67±.01 +.07	.36±.01 +.01	.25±.01 -.02	.21±.01 -.03	.67±.01 +.01	.36±.02 -.02	.23±.01 +.00	.19±.02 -.01
	Last Answer	.74±.01 -.07	.48±.02 -.10	.22±.03 +.05	.20±.05 +.08	.74±.02 -.04	.48±.02 -.09	.19±.02 +.02	.15±.03 +.02
pl	Last Query	.69±.01 +.01	.42±.01 +.00	.24±.01 +.01	.21±.02 +.02	.64±.02 -.06	.27±.02 -.09	.27±.05 +.09	.26±.06 +.10
	Last Answer	.72±.01 -.07	.50±.02 -.06	.40±.04 +.22	.41±.05 +.28	.71±.02 -.05	.38±.03 -.08	.16±.01 +.00	.14±.02 +.00
ru	Last Query	.69±.01 +.03	.33±.01 +.00	.28±.02 +.09	.28±.02 +.10	.65±.01 -.11	.26±.01 -.19	.22±.02 +.05	.20±.03 +.06
	Last Answer	.74±.01 -.09	.40±.02 -.14	.47±.04 +.33	.52±.04 +.41	.73±.01 -.06	.42±.02 -.11	.15±.01 -.01	.11±.02 -.02
ja	Last Query	.64±.01 -.09	.24±.01 -.05	.59±.07 +.43	.65±.06 +.51	.62±.02 -.02	.23±.02 -.03	.52±.08 +.33	.55±.08 +.38
	Last Answer	.64±.01 -.16	.32±.03 -.10	.67±.03 +.54	.68±.03 +.58	.71±.02 -.02	.33±.04 -.08	.26±.05 +.09	.27±.07 +.11

Table 6: Results for Llama 3.1 8B on MKQA and Global-MMLU, with English as the source language. First group shows the results for the probe and baselines trained and tested on French. all* = avg. across all languages. Remaining groups list zero-shot performance on test languages. Last Query / Last Answer = Probe trained on last query / answer hiddens. Gray numbers state performance difference compared to the oracle probe, i.e. $\Delta < 0$ indicating that the oracle scored higher.

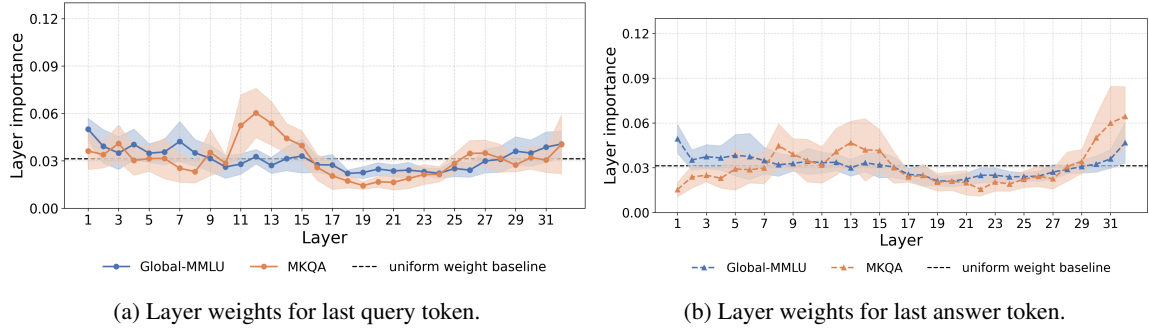


Figure 9: Distribution of learned layer weights from probes trained with different random seeds on the hidden states of the last (a) query token and (b) answer token for Llama 3.1 8B on Global-MMLU and MKQA. Shaded areas indicate ± 1 standard deviation, dashed line shows uniform weight.

Weight Ablation. We selectively make layers unavailable to the probe and quantify the impact of this change on the uncertainty estimation task using a sliding window approach. Recall the computation of layer weights in Equation (2). To ablate a window W of k contiguous layers, we intervene on the probe’s layer-mixing logits by setting $\{w_\ell : \ell \in W\}$ to $-\infty$:

$$\alpha_\ell = \text{softmax}(\mathbf{w}')_\ell = \frac{\exp(w'_\ell)}{\sum_{k=1}^L \exp(w'_k)}, \quad (6)$$

where

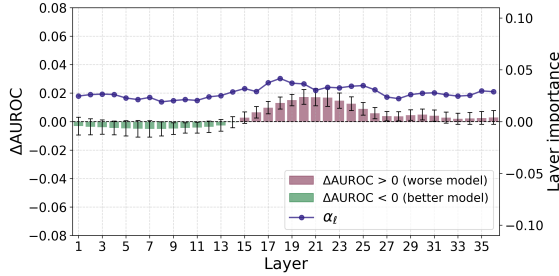
$$w'_\ell = \begin{cases} -\infty, & \ell \in W \\ w_\ell, & \ell \notin W. \end{cases} \quad (7)$$

Representation Ablation. Recall that the probe computes a logit as $z_i = \sum_{\ell=1}^L \alpha_\ell s_{i,\ell} + b$ (Equations (3) and (4)), where $s_{i,\ell} = \mathbf{v}^\top \mathbf{h}_i^{(\ell)}$ denoted here. To ablate a window W of k contiguous layers, for each example i , we switch its score $s_{i,\ell}$ with the one of another example $s_{j,\ell}$, selected by independently sampling it from the test set. This is done to remove useful information for a specific test sample, but to not introduce corrupting information that might unfairly distort probe performance. The logit of an example i after ablation of window W is therefore:

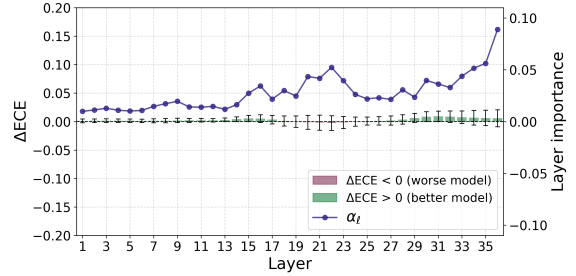
$$z_i^{(W)} = \sum_{\ell \notin W} \alpha_\ell s_{i,\ell} + \sum_{\ell \in W} \alpha_\ell \tilde{s}_{i,\ell} + b, \quad (8)$$

		MKQA				MMLU			
		AUROC↑	AUPR↑	Brier↓	ECE↓	AUROC↑	AUPR↑	Brier↓	ECE↓
fr	Last Query	.70±.01	.44±.01	.23±.01	.18±.01	.71±.01	.60±.01	.25±.01	.19±.01
	Last Answer	.78±.01	.57±.01	.18±.01	.13±.01	.76±.01	.69±.01	.23±.01	.19±.01
fr	Last Query	.68±.01 -.05	.31±.01 -.05	.18±.01 +.01	.15±.01 +.01	.66±.01 -.05	.41±.01 -.08	.25±.03 +.04	.20±.05 +.04
	Last Answer	.77±.01 -.05	.44±.02 -.05	.16±.02 +.02	.14±.03 +.02	.80±.01 +.02	.63±.02 +.05	.20±.01 +.01	.18±.02 +.02
es	Last Query	.71±.01 -.01	.32±.02 -.04	.19±.01 +.01	.16±.01 +.00	.71±.01 +.03	.44±.01 +.00	.23±.02 +.01	.19±.03 +.01
	Last Answer	.79±.01 -.04	.47±.02 -.05	.23±.05 +.09	.25±.07 +.14	.77±.01 +.00	.61±.01 +.06	.22±.02 +.02	.21±.03 +.04
pl	Last Query	.66±.01 -.07	.33±.01 -.01	.18±.01 +.02	.16±.02 +.02	.66±.01 -.06	.33±.01 -.09	.21±.01 +.00	.20±.01 +.03
	Last Answer	.74±.01 -.08	.47±.01 -.01	.20±.02 +.07	.21±.03 +.10	.76±.01 -.03	.46±.02 -.06	.26±.03 +.08	.27±.03 +.11
ru	Last Query	.70±.01 +.01	.25±.02 -.03	.14±.02 -.02	.14±.02 -.02	.64±.01 -.09	.31±.01 -.15	.22±.01 +.03	.19±.01 +.03
	Last Answer	.70±.02 -.09	.36±.02 -.07	.21±.03 +.09	.25±.04 +.14	.77±.01 -.03	.52±.01 -.04	.17±.01 +.00	.15±.01 +.01
ja	Last Query	.61±.03 -.17	.16±.01 -.17	.18±.03 +.05	.17±.05 +.06	.65±.02 -.09	.32±.02 -.13	.20±.03 +.01	.15±.06 -.02
	Last Answer	.78±.01 -.09	.35±.02 -.18	.19±.03 +.10	.20±.04 +.13	.76±.01 -.03	.45±.03 -.07	.23±.03 +.05	.22±.03 +.06

Table 7: Results for Qwen 3 8B on MKQA and Global-MMLU, with English as the source language. First group shows the results for the probe and baselines trained and tested on French. all* = avg. across all languages. Remaining groups list zero-shot performance on test languages. Last Query / Last Answer = Probe trained on last query / answer hiddens. Gray numbers state performance difference compared to the oracle probe, i.e. $\Delta < 0$ indicating that the oracle scored higher.



(a) Δ in AUROC on MKQA.



(b) Δ in ECE score on Global-MMLU.

Figure 10: Impact of layer ablation on zero-shot AUROC and ECE (avg. over all languages) for Qwen3 8B. Bars show mean change in metric value over 10 random seeds. Whiskers denote ± 1 standard deviation.

with

$$s_{i,\ell} = \mathbf{v}^\top \mathbf{h}_i^{(\ell)}, \quad \tilde{s}_{i,\ell} = s_{\pi(i),\ell}, \quad (9)$$

where $\pi(\cdot)$ a permutation function defined on the natural numbers up to the cardinality of the test set, mapping each index to another index sampling uniformly without replacement and independently for each layer.

Results. Given the volume of results, we show only a selection here: the representation-ablation counterpart to Figure 6 in Figure 10, and complementary Llama 3.1 results in Figure 11.

A.5 Uniform Probe Weights

We conduct additional experiments fixing the layer weights in Equation (2) to uniform and only learning the linear layer parameters. We show the per-

formance differences for both model in Figures 12 and 13. The comparison remains overall inconsistent, but we attribute the fact that the uniform probe is able to achieve comparable results to two factors: First of all, the informational redundancy in transformer layers helps to preserve information in the pooled representation even when weights are uniform, and additionally the large amount of parameters in the linear layer (4096 given the size of hidden representations) helps to learn useful features for the task. Nevertheless, we see our results and works such as Zhou et al. (2025) as evidence that future, more robust works will likely rely on specific layers.

		MKQA				Global-MMLU			
		AUROC↑	AUPR↑	Brier↓	ECE↓	AUROC↑	AUPR↑	Brier↓	ECE↓
fr	Seq. Likelihood	.80±.00	.45±.00	.52±.00	.62±.00	.72±.00	.53±.00	.51±.00	.58±.00
	P(True)	.63±.00	.22±.00	.18±.00	.18±.00	.55±.00	.32±.00	.27±.00	.27±.00
	Mass-Mean Probe	.67±.00	.52±.00	.43±.00	.43±.00	.60±.00	.52±.00	.40±.00	.40±.00
	Verbalized Conf.	.71±.00	.44±.00	.64±.00	.70±.00	.75±.00	.59±.00	.54±.00	.59±.00
	Cross-ling. Probe	.82±.01	.49±.02	.14±.01	.12±.01	.78±.00	.58±.01	.19±.00	.16±.00
en	Seq. Likelihood	.80±.00	.61±.00	.45±.00	.54±.00	.73±.00	.66±.00	.45±.00	.48±.00
	P(True)	.69±.00	.42±.00	.26±.00	.26±.00	.62±.00	.48±.00	.39±.00	.39±.00
	Mass-Mean Probe	.67±.00	.58±.00	.36±.00	.37±.00	.56±.00	.64±.00	.48±.00	.48±.00
	Verbalized Conf.	.70±.00	.57±.00	.61±.00	.64±.00	.72±.00	.68±.00	.51±.00	.54±.00
	Cross-ling. Probe	.70±.01	.44±.01	.24±.01	.22±.01	.72±.01	.60±.01	.27±.01	.24±.01
es	Seq. Likelihood	.81±.00	.52±.00	.51±.00	.61±.00	.70±.00	.51±.00	.53±.00	.59±.00
	P(True)	.30±.00	.13±.00	.19±.00	.19±.00	.41±.00	.23±.00	.26±.00	.26±.00
	Mass-Mean Probe	.65±.00	.52±.00	.45±.00	.45±.00	.59±.00	.50±.00	.41±.00	.41±.00
	Verbalized Conf.	.75±.00	.51±.00	.54±.00	.59±.00	.73±.00	.55±.00	.52±.00	.58±.00
	Cross-ling. Probe	.82±.01	.53±.03	.19±.02	.20±.03	.74±.01	.52±.01	.23±.01	.21±.02
pl	Seq. Likelihood	.82±.00	.48±.00	.44±.00	.57±.00	.79±.00	.53±.00	.48±.00	.58±.00
	P(True)	.61±.00	.19±.00	.15±.00	.15±.00	.53±.00	.26±.00	.21±.00	.21±.00
	Mass-Mean Probe	.71±.00	.50±.00	.34±.00	.35±.00	.59±.00	.58±.00	.57±.00	.57±.00
	Verbalized Conf.	.67±.00	.40±.00	.60±.00	.65±.00	.76±.00	.52±.00	.53±.00	.59±.00
	Cross-ling. Probe	.73±.01	.35±.03	.23±.03	.24±.03	.75±.01	.48±.01	.27±.03	.27±.03
ru	Seq. Likelihood	.79±.00	.35±.00	.49±.00	.62±.00	.75±.00	.47±.00	.54±.00	.62±.00
	P(True)	.43±.00	.12±.00	.12±.00	.12±.00	.45±.00	.21±.00	.21±.00	.21±.00
	Mass-Mean Probe	.73±.00	.51±.00	.40±.00	.41±.00	.62±.00	.57±.00	.51±.00	.51±.00
	Verbalized Conf.	.72±.00	.36±.00	.55±.00	.61±.00	.73±.00	.49±.00	.51±.00	.57±.00
	Cross-ling. Probe	.80±.01	.43±.03	.15±.03	.16±.04	.72±.01	.43±.01	.39±.03	.42±.03
ja	Seq. Likelihood	.81±.00	.41±.00	.43±.00	.56±.00	.74±.00	.49±.00	.48±.00	.57±.00
	P(True)	.60±.00	.16±.00	.13±.00	.13±.00	.54±.00	.24±.00	.21±.00	.21±.00
	Mass-Mean Probe	.83±.00	.57±.00	.22±.00	.23±.00	.52±.00	.59±.00	.73±.00	.73±.00
	Verbalized Conf.	.67±.00	.37±.00	.66±.00	.72±.00	.72±.00	.44±.00	.57±.00	.63±.00
	Cross-ling. Probe	.64±.04	.23±.04	.16±.02	.16±.02	.72±.01	.44±.02	.31±.02	.31±.02

Table 8: Complete answer-token confidence estimation comparison for Qwen 3 8B, with probes trained in French.

		MKQA				Global-MMLU			
		AUROC↑	AUPR↑	Brier↓	ECE↓	AUROC↑	AUPR↑	Brier↓	ECE↓
fr	Seq. Likelihood	.72±.00	.51±.00	.39±.00	.45±.00	.77±.00	.57±.00	.28±.00	.35±.00
	P(True)	.55±.00	.34±.00	.27±.00	.27±.00	.53±.00	.27±.00	.24±.00	.24±.00
	Mass-Mean Probe	.61±.00	.57±.00	.48±.00	.48±.00	.64±.00	.56±.00	.46±.00	.46±.00
	Verbalized Conf.	.62±.00	.61±.00	.55±.00	.57±.00	.67±.00	.60±.00	.58±.00	.62±.00
	Cross-ling. Probe	.82±.00	.58±.01	.17±.00	.11±.01	.76±.01	.51±.03	.19±.01	.15±.01
en	Seq. Likelihood	.72±.00	.59±.00	.32±.00	.33±.00	.80±.00	.71±.00	.26±.00	.28±.00
	P(True)	.35±.00	.30±.00	.39±.00	.39±.00	.51±.00	.35±.00	.36±.00	.36±.00
	Mass-Mean Probe	.59±.00	.65±.00	.47±.00	.47±.00	.68±.00	.67±.00	.38±.00	.38±.00
	Verbalized Conf.	.61±.00	.67±.00	.50±.00	.51±.00	.74±.00	.69±.00	.43±.00	.47±.00
	Cross-ling. Probe	.76±.00	.62±.01	.23±.01	.16±.03	.75±.01	.61±.02	.22±.01	.17±.03
es	Seq. Likelihood	.73±.00	.44±.00	.38±.00	.46±.00	.77±.00	.58±.00	.29±.00	.36±.00
	P(True)	.45±.00	.21±.00	.24±.00	.24±.00	.56±.00	.27±.00	.23±.00	.23±.00
	Mass-Mean Probe	.57±.00	.55±.00	.56±.00	.56±.00	.66±.00	.58±.00	.46±.00	.46±.00
	Verbalized Conf.	.67±.00	.60±.00	.51±.00	.53±.00	.73±.00	.61±.00	.55±.00	.61±.00
	Cross-ling. Probe	.77±.01	.48±.01	.50±.05	.54±.04	.78±.01	.53±.03	.16±.01	.12±.02
pl	Seq. Likelihood	.75±.00	.52±.00	.31±.00	.38±.00	.80±.00	.56±.00	.31±.00	.42±.00
	P(True)	.63±.00	.32±.00	.25±.00	.25±.00	.48±.00	.18±.00	.19±.00	.19±.00
	Mass-Mean Probe	.65±.00	.49±.00	.31±.00	.31±.00	.67±.00	.50±.00	.36±.00	.36±.00
	Verbalized Conf.	.61±.00	.56±.00	.52±.00	.54±.00	.66±.00	.48±.00	.51±.00	.58±.00
	Cross-ling. Probe	.74±.01	.47±.01	.19±.01	.15±.01	.77±.01	.47±.02	.32±.07	.36±.08
ru	Seq. Likelihood	.75±.00	.40±.00	.35±.00	.47±.00	.78±.00	.53±.00	.29±.00	.39±.00
	P(True)	.64±.00	.24±.00	.14±.00	.06±.00	.55±.00	.26±.00	.16±.00	.10±.00
	Mass-Mean Probe	.66±.00	.45±.00	.35±.00	.35±.00	.61±.00	.48±.00	.47±.00	.47±.00
	Verbalized Conf.	.60±.00	.49±.00	.59±.00	.60±.00	.63±.00	.53±.00	.61±.00	.65±.00
	Cross-ling. Probe	.78±.01	.41±.01	.14±.01	.11±.03	.73±.01	.48±.02	.31±.07	.35±.08
ja	Seq. Likelihood	.76±.00	.32±.00	.35±.00	.48±.00	.78±.00	.53±.00	.26±.00	.35±.00
	P(True)	.53±.00	.13±.00	.12±.00	.11±.00	.61±.00	.24±.00	.18±.00	.17±.00
	Mass-Mean Probe	.62±.00	.41±.00	.42±.00	.42±.00	.69±.00	.42±.00	.24±.00	.24±.00
	Verbalized Conf.	.70±.00	.37±.00	.43±.00	.55±.00	.61±.00	.38±.00	.47±.00	.55±.00
	Cross-ling. Probe	.77±.01	.32±.01	.43±.10	.49±.10	.72±.01	.43±.03	.28±.04	.29±.04

Table 9: Complete answer-token confidence estimation comparison for Llama 3.1 8B, with probes trained in French.

		MKQA				Global-MMLU			
		AUROC↑	AUPR↑	Brier↓	ECE↓	AUROC↑	AUPR↑	Brier↓	ECE↓
fr	Seq. Likelihood	.63±.00	.26±.00	.15±.00	.12±.00	.50±.00	.26±.00	.24±.00	.19±.00
	Mass-Mean Probe	.64±.00	.47±.00	.44±.00	.44±.00	.69±.00	.57±.00	.38±.00	.38±.00
	Cross-ling. Probe	.73±.01	.36±.01	.17±.01	.14±.00	.71±.01	.49±.02	.21±.01	.16±.01
en	Seq. Likelihood	.62±.00	.38±.00	.24±.00	.22±.00	.51±.00	.38±.00	.36±.00	.34±.00
	Mass-Mean Probe	.69±.00	.59±.00	.41±.00	.42±.00	.65±.00	.64±.00	.40±.00	.40±.00
	Cross-ling. Probe	.70±.01	.47±.01	.22±.01	.18±.01	.64±.02	.51±.02	.31±.01	.26±.01
es	Seq. Likelihood	.64±.00	.28±.00	.16±.00	.12±.00	.52±.00	.26±.00	.22±.00	.18±.00
	Mass-Mean Probe	.64±.00	.49±.00	.43±.00	.44±.00	.69±.00	.56±.00	.37±.00	.37±.00
	Cross-ling. Probe	.71±.01	.34±.01	.17±.00	.14±.01	.68±.01	.41±.02	.27±.02	.25±.03
pl	Seq. Likelihood	.63±.00	.26±.00	.13±.00	.09±.00	.57±.00	.26±.00	.18±.00	.12±.00
	Mass-Mean Probe	.67±.00	.46±.00	.41±.00	.41±.00	.72±.00	.52±.00	.33±.00	.33±.00
	Cross-ling. Probe	.67±.01	.28±.01	.15±.01	.13±.01	.73±.01	.40±.01	.20±.01	.17±.02
ru	Seq. Likelihood	.65±.00	.19±.00	.11±.00	.06±.00	.51±.00	.24±.00	.18±.00	.12±.00
	Mass-Mean Probe	.69±.00	.46±.00	.40±.00	.41±.00	.68±.00	.50±.00	.39±.00	.39±.00
	Cross-ling. Probe	.72±.01	.30±.01	.12±.00	.11±.01	.68±.01	.35±.02	.23±.02	.21±.03
ja	Seq. Likelihood	.59±.00	.18±.00	.12±.00	.08±.00	.52±.00	.23±.00	.19±.00	.16±.00
	Mass-Mean Probe	.56±.00	.56±.00	.85±.00	.85±.00	.67±.00	.52±.00	.43±.00	.43±.00
	Cross-ling. Probe	.57±.02	.16±.01	.13±.01	.11±.01	.64±.01	.32±.02	.20±.02	.17±.03

Table 10: Complete query-token confidence estimation comparison for Qwen 3 8B, with probes trained in French.

		MKQA				Global-MMLU			
		AUROC↑	AUPR↑	Brier↓	ECE↓	AUROC↑	AUPR↑	Brier↓	ECE↓
fr	Seq. Likelihood	.66±.00	.40±.00	.23±.00	.20±.00	.52±.00	.26±.00	.20±.00	.14±.00
	Mass-Mean Probe	.63±.00	.50±.00	.41±.00	.41±.00	.71±.00	.56±.00	.33±.00	.33±.00
	Cross-ling. Probe	.68±.00	.45±.00	.24±.00	.20±.01	.73±.01	.48±.02	.21±.00	.17±.01
en	Seq. Likelihood	.62±.00	.47±.00	.37±.00	.36±.00	.46±.00	.32±.00	.33±.00	.31±.00
	Mass-Mean Probe	.65±.00	.60±.00	.39±.00	.39±.00	.73±.00	.68±.00	.32±.00	.32±.00
	Cross-ling. Probe	.67±.01	.53±.01	.28±.00	.24±.01	.70±.02	.57±.02	.26±.01	.22±.01
es	Seq. Likelihood	.60±.00	.31±.00	.21±.00	.18±.00	.57±.00	.30±.00	.19±.00	.13±.00
	Mass-Mean Probe	.66±.00	.51±.00	.42±.00	.43±.00	.72±.00	.57±.00	.33±.00	.33±.00
	Cross-ling. Probe	.64±.00	.35±.01	.24±.01	.20±.01	.71±.02	.45±.02	.23±.02	.21±.03
pl	Seq. Likelihood	.59±.00	.35±.00	.22±.00	.20±.00	.53±.00	.19±.00	.16±.00	.08±.00
	Mass-Mean Probe	.69±.00	.53±.00	.36±.00	.36±.00	.73±.00	.51±.00	.30±.00	.30±.00
	Cross-ling. Probe	.68±.01	.39±.01	.22±.00	.18±.01	.69±.01	.39±.02	.21±.03	.20±.04
ru	Seq. Likelihood	.63±.00	.22±.00	.14±.00	.11±.00	.59±.00	.25±.00	.16±.00	.10±.00
	Mass-Mean Probe	.71±.00	.44±.00	.34±.00	.34±.00	.66±.00	.48±.00	.37±.00	.38±.00
	Cross-ling. Probe	.71±.01	.38±.01	.19±.01	.18±.01	.67±.01	.34±.01	.23±.02	.21±.03
ja	Seq. Likelihood	.58±.00	.16±.00	.11±.00	.07±.00	.60±.00	.24±.00	.16±.00	.11±.00
	Mass-Mean Probe	.69±.00	.32±.00	.53±.00	.55±.00	.65±.00	.50±.00	.46±.00	.46±.00
	Cross-ling. Probe	.64±.01	.21±.01	.21±.04	.22±.05	.69±.02	.32±.02	.20±.03	.18±.04

Table 11: Complete query-token confidence estimation comparison for Llama 3.1 8B, with probes trained in French.

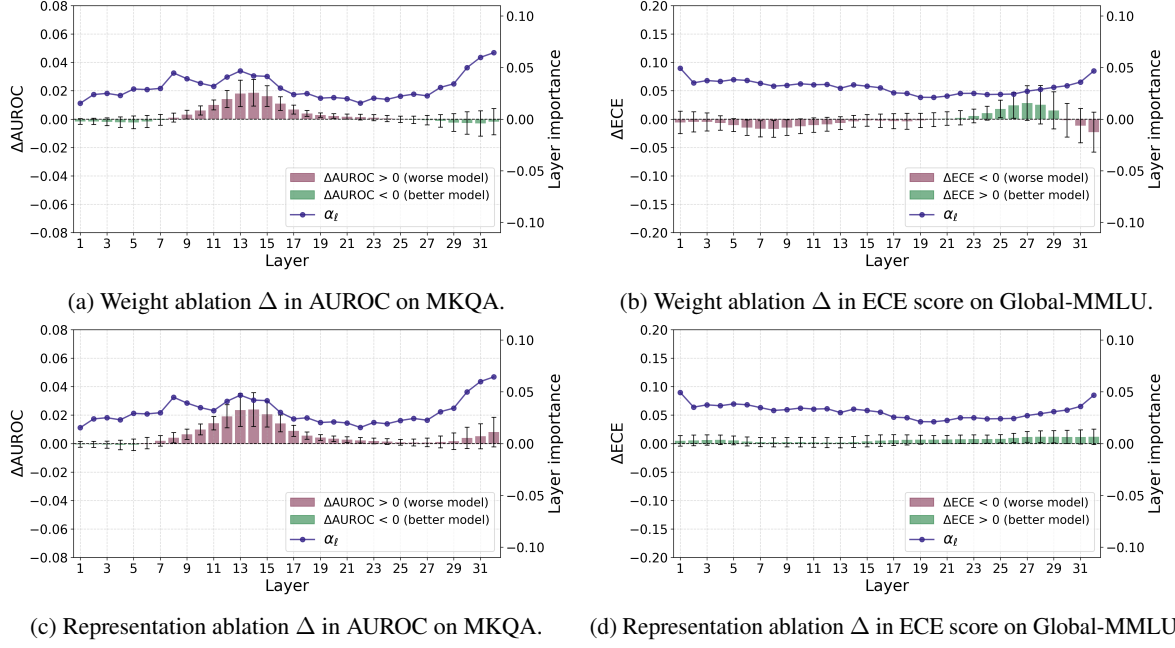
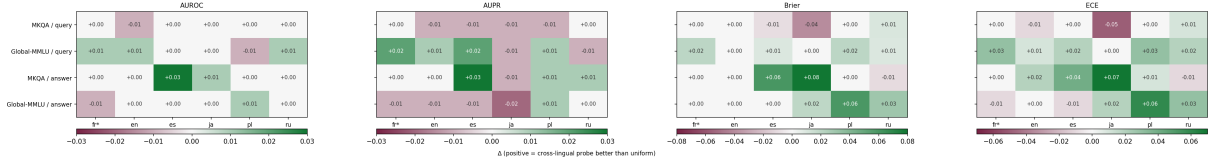
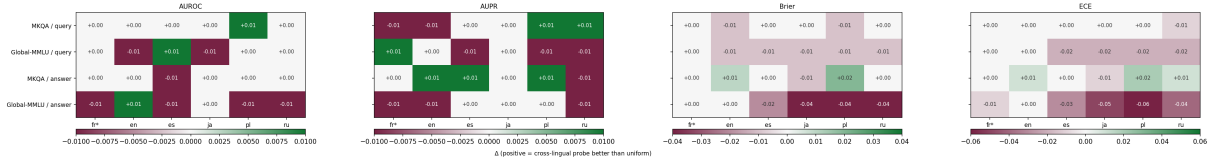


Figure 11: Impact of layer ablation on zero-shot AUROC and ECE (avg. over all languages) for Llama 3.1 8B. Bars show mean change in metric value over 10 random seeds. Whiskers denote ± 1 standard deviation.



B Reproducibility Details

B.1 Data Pre-Processing

In the following, we describe the exact preprocessing used for MKQA and Global-MMLU. The final number of examples per dataset, model and language can be found in Table 12.

B.1.1 MKQA

The MKQA dataset (Longpre et al., 2021) (GitHub repository) comprises 10000 questions parallel across 26 languages, each annotated with one or more answer types (binary, date, entity, long answer, number, number with unit, short phrase, and unanswerable). We retain only questions of type date, entity, number, number with unit, and short phrase. Binary questions are excluded due to limited token generation, while long-answer and unanswerable questions are excluded because no ground-truth answers are provided in the dataset. Duplicate questions are also removed.

We further exclude questions with time-sensitive ground-truth answers that are no longer reliable. To identify such cases, we employ an LLM-as-a-judge approach using GPT-4.1 with the prompt shown in Figure 14. We use default parameters, except for a temperature of 0 and a maximum of 5 generated tokens. Filtering is performed on English questions and then propagated to other languages. As a result, the final dataset contains 5237 questions per language (date: 943, entity: 3347, number: 258, number with unit: 254, short phrase: 435).

B.1.2 Global-MMLU

For our experiments, we use the test split of the Global-MMLU dataset (Singh et al., 2024).⁸ The dataset spans 42 languages with 14042 multiple-choice questions across 57 subjects. Following Chandak et al. (2025), we identify and retain only those questions that are sufficiently self-contained to be posed in an open-ended format, using the following procedure applied to the English questions.

We first apply a RegEx-based filtering step to remove questions that are inherently tied to the multiple-choice format. Specifically, we remove questions that (1) contain keywords indicative of multiple-choice framing, such as "following", "which of these combinations", or "options", (2) contain fill-in-the-blank placeholders (e.g., three or more consecutive underscores), (3) reference

⁸<https://huggingface.co/datasets/CohereLabs/Global-MMLU>

Dataset	LLM	en	es	fr	ja	pl	ru
MKQA	Llama 3.1 8B	5216	5219	5205	5176	5092	5106
	Qwen3 8B	5225	5226	5222	5178	5186	5162
Global-MMLU	Llama 3.1 8B	4077	4148	4067	4178	4111	4111
	Qwen3 8B	4212	4198	4186	4195	4120	4178

Table 12: Total number of examples per dataset, model, and language.

"Statement 1" and "Statement 2" simultaneously, as these questions require evaluating paired assertions, or (4) ask the model to identify a true or false statement, as indicated by truth-value keywords ("true", "false", "correct", "incorrect"), unless the question also contains the word "following", in which case it is already caught by criterion (1).

We further use GPT-4.1 to score how well each question can be answered without the multiple-choice options, on a scale from 1 to 10 (see Figure 15 for the full prompt). Table 13 outlines the score distribution across the questions. We retain only questions with a score of 8 or above, as these are sufficiently self-contained to be posed as open-ended questions. After these preprocessing steps we are left with 4215 questions. For all other languages, we retain the translations corresponding to these questions. Note that with this procedure we do not rephrase any question, ensuring that no additional wording bias is introduced during conversion to open-ended format.

Score	Count	Percentage
1	43	0.58%
2	50	0.67%
3	342	4.61%
4	287	3.87%
5	1171	15.79%
6	1004	13.54%
7	303	4.09%
8	2291	30.90%
9	851	11.48%
10	1073	14.47%

Table 13: GPT-4.1 score distribution for question answerability without multiple-choice options.

B.1.3 Normalization & Splitting

(Sun et al., 2025) showed that in transformer-based LLMs using pre-layer normalization (e.g., Llama), hidden-state variance grows exponentially with depth, hindering deeper-layer training. To address

Dataset	Question Type	Prompt
MKQA	short phrase	P_3
	entity	P_3
	date	P_4
	number	P_5
	number with unit	P_6
Global MMLU	(all)	P_7

Table 14: Mapping of evaluation splits to prompt templates.

this and ensure that probe weights capture informative signal rather than magnitude differences, we apply per-layer z-score standardization with scikit-learn’s StandardScaler.

For each feature x , we compute the standardized value $x' = (x - \mu)/\sigma$, where μ and σ are the training-set mean and standard deviation, centering features to zero mean and unit variance. The fitted scaler is applied unchanged to the validation and test splits of the training language, and to the test sets of all zero-shot languages. We do not further normalize, as residual magnitude differences may be informative.

Given that both datasets are parallel and class ratios vary by language, we stratify splits to prevent information leakage and ensure balanced evaluation. For each training language, hidden states are divided into train/validation/test sets (0.7:0.15:0.15) while preserving correct/incorrect ratios. Evaluation samples are drawn from the test sets so that these ratios match across all languages.

B.2 LLM Prompting

The system instructions used to prompt the considered LLMs across languages are in Table 14.

B.3 Answer Correctness Assessment

We use GPT-4.1 to assess answer correctness, with the prompt shown in Figure 17. For MKQA, GPT-4.1 was used across all languages and both models. For Global-MMLU, due to resource constraints we used GPT-4.1 for French and GPT-4.1-mini for the remaining languages (for both Llama 3.1 8B and Qwen3 8B). To verify that this substitution does not compromise label quality, we measured inter-judge agreement between GPT-4.1 and GPT-4.1-mini on Global-MMLU, for Llama 3.1 8B, in French, obtaining 94.4% agreement and a Cohen’s κ of 0.849.

B.4 Confidence Probe Implementation Details

The layer weights $\mathbf{w}$ were initialized to zero (uniform weighting across layers), the hidden-dimension weights $\mathbf{v}$ were sampled from the normal distribution $\mathcal{N}(0, 1)$ (to break symmetry and enable diverse feature weighting), and the bias b was set to zero (no offset prior to training). Probes were trained with binary cross-entropy loss with logits and L_2 regularization for $\mathbf{v}$:

$$\mathcal{L}(\mathbf{y}, \mathbf{z}) = -\frac{1}{N} \sum_{i=1}^N \left[w_+ y_i \log \sigma(z_i) + (1 - y_i) \log(1 - \sigma(z_i)) \right] + \lambda \|\mathbf{v}\|_2^2 \quad (10)$$

where N is the number of training examples, $y_i \in \{0, 1\}$ is the ground-truth correctness label, $z_i \in \mathbb{R}$ is the logit produced by the probe, and $\sigma(\cdot)$ is the sigmoid function. The parameter $w_+ \in \mathbb{R}^+$ is the correct-class weight to address class imbalance, computed from the ratio of negative to positive examples in the training set.⁹ $\lambda \in \mathbb{R}^+$ is the L_2 regularisation applied only to $\mathbf{v}$, with no regularisation applied to $\mathbf{w}$ or b .

B.5 Hyperparameter Tuning

We tuned hyperparameters with Ray Tune using the ASHA scheduler for inter-trial early stopping, optimizing the scalar objective val_loss (minimized). Each trial sampled from a log-uniform search space: learning rate $1r \in [10^{-5}, 10^{-1}]$ and L2 regularization on the neuron weights $\mathbf{v}$, $l2_lambda \in [10^{-4}, 3 \times 10^{-1}]$. The batch size and maximum epochs were fixed at 4 and 1000, respectively. During each trial, we trained with AdamW using selective weight decay (L2 applied only to $\mathbf{v}$; none for layer weights $\mathbf{w}$ or bias terms), and reported the mean training loss and validation loss at every epoch to Ray Tune. Validation loss was computed on a fixed held-out validation split. To address class imbalance, we used BCEWithLogitsLoss with a positive-class weight proportional to the ratio of negative to positive examples in the training set. Trials were stopped early by ASHA (max_t=1000, grace_period=30, reduction_factor=2) when underperforming, while in-trial early stopping with patience = 20 halted training if val_loss failed to improve. After 50 trials, Ray Tune selected the best

⁹For more information, see the pos_weight parameter of PyTorch’s BCEWithLogitsLoss class.

configuration according to the minimum validation loss. The selected hyperparameters are reported in Table 15.

Dataset	LLM	Probe	Learning rate	Epochs	L2 regularisation for v
MKQA	Llama 3.1 8B	Query	0.0008	139	0.2861
		Answer	0.0005	210	0.2728
	Qwen3 8B	Query	0.0009	98	0.2953
		Answer	0.0009	111	0.2225
Global-MMLU	Llama 3.1 8B	Query	0.0011	94	0.2826
		Answer	0.0020	60	0.2557
	Qwen3 8B	Query	0.0010	103	0.2687
		Answer	0.0010	165	0.2115

Table 15: Selected hyperparameters for probes trained on last-token query and answer representations for MKQA and Global-MMLU using Llama 3.1 8B and Qwen3 8B models.

P₁ - MKQA Time-Sensitivity Filtering Prompt (English only)

Your task is to determine whether the following question is TIME-SENSITIVE relative to when the dataset was collected (around 2019). Assume the question was asked in or before 2019.

A question is TIME-SENSITIVE if its correct answer depends on the real-world time at which the question is asked. In particular, a question is time-sensitive if its answer depends on:

- The current date, year, or recent events,
- Relative time expressions (e.g., “this year”, “currently”, “latest”, “now”, “recent”),
- The current real-world status, role, ranking, statistic, population, record, or version,
- Ongoing or changing real-world situations.

A question is NOT time-sensitive if its answer is fixed by historical facts, completed events, or already-released creative works (such as movies, TV series, books, or games), and does not change depending on when the question is asked.

Output Instructions: Answer with exactly one of the following:

- YES - The question is time-sensitive (its answer would differ depending on when it is asked)
- NO - The question is not time-sensitive (its answer would be the same regardless of whether it is asked in 2015, 2019, or 2024)
- UNCERTAIN - The question cannot be reasonably judged

Do not provide any explanation.

Examples

Time-sensitive (YES):

- who is the prime minister of japan
- what is the population of india
- who is the ceo of apple
- what is the latest version of android
- what was the winner of last year’s champions league
- how old is the president of france

Not time-sensitive (NO):

- who wrote the novel 1984
- when was the berlin wall built
- who discovered penicillin
- what is the capital of argentina
- when does prentis come back to criminal minds
- who painted the sistine chapel ceiling

Question: {question}

Figure 14: Prompt used by the LLM-as-a-judge to filter time-sensitive questions from MKQA, whose ground-truth answers may no longer be reliable. Applied to the English questions only.

P₂ - Answerability Scoring Prompt (English only)

Your task is to evaluate how well a multiple-choice question or incomplete statement (stem) can be answered without seeing the answer choices.

Assign a score from 1 to 10 based only on whether the stem can be answered or completed without the choices.

Scoring Scale (1-10):

Score 1-2: The stem does not define a clear task and cannot be answered without choices. Examples:

- "Oxygen is used:"
- "The element (4, 2) of"

Score 3-4: The stem defines a task, but answering it requires choosing from an unspecified list or comparison that is only provided by the choices. Examples:

- "Which set of integers is in order from least to greatest?"
- "The most accurate statement is"

Score 5: The stem can be answered without choices, but allows many reasonable completions or answers. Examples:

- "Under which circumstances would you not use a catheter valve?"
- "The advantages of X-rays include"

Score 6-7: The stem can be answered without choices and limits the topic, but allows variation in the expected answer. Examples:

- "What is one function of the liver?"
- "As temperature increases, reaction rate generally"

Score 8-9: The stem can be answered without choices and strongly limits the expected answer. Examples:

- "What gas do plants primarily absorb during photosynthesis?"
- "The enzymes of glycolysis are located in the"

Score 10: The stem alone specifies a single, exact expected answer. Examples:

- "What is the first law of thermodynamics in physics?"
- "Calculate 15×12 ."

Output Instructions:

You must output exactly one line containing only one integer from 1 to 10, without any additional text.

Response Format (must match exactly):

SCORE: X

Figure 15: System prompt used by the LLM-as-a-judge to assess stem answerability on a 1-10 scale.

P₃ - MKQA: short phrase / entity questions (6 languages)

- en** You are a helpful assistant. Answer the user's question in one to ten words. End your output immediately after your answer to the question.
- fr** Tu es un assistant serviable. Réponds à la question de l'utilisateur en un à dix mots. Arrête la génération immédiatement après ta réponse à la question.
- es** Eres un asistente útil. Responde la pregunta del usuario en una a diez palabras. Detén la generación inmediatamente después de responder la pregunta.
- pl** Jesteś użytecznym asystentem. Odpowiadaj na pytania użytkownika, używając od jednego do dziesięciu słów. Przestań generować natychmiast po udzieleniu odpowiedzi.
- ru** Ты полезный помощник. Отвечай на вопросы пользователя, используя от одного до десяти слов. Останавливай генерацию сразу после ответа.
- ja** あなたは役に立つアシスタントです。ユーザーの質問に1～10語で答えてください。質問に答えたらすぐに出力を終了してください。

P₄ - MKQA: date questions (6 languages)

- en** You are a helpful assistant. Answer the user's question using only the date requested. Do not include explanations or extra text. End your output immediately after your answer to the question.
- fr** Tu es un assistant serviable. Réponds à la question de l'utilisateur en utilisant uniquement la date demandée. N'ajoute ni explications ni texte supplémentaire. Arrête la génération immédiatement après ta réponse à la question.
- es** Eres un asistente útil. Responde la pregunta del usuario usando solo la fecha solicitada. No incluyas explicaciones ni texto adicional. Detén la generación inmediatamente después de responder la pregunta.
- pl** Jesteś użytecznym asystentem. Odpowiadaj na pytania użytkownika, używając wyłącznie żądanej daty. Nie dodawaj wyjaśnień ani dodatkowego tekstu. Przestań generować natychmiast po udzieleniu odpowiedzi.
- ru** Ты полезный помощник. Отвечай на вопросы пользователя, используя только указанную дату. Не добавляй пояснений или дополнительного текста. Останавливай генерацию сразу после ответа.
- ja** あなたは役に立つアシスタントです。ユーザーの質問には、要求された日付のみを使って回答してください。説明や余分なテキストは含めないでください。質問への回答後、すぐに出力を終了してください。

P₅ - MKQA: number questions (6 languages)

- en** You are a helpful assistant. Answer the user's question using only the number needed to answer it. Do not include explanations or extra text. End your output immediately after your answer to the question.
- fr** Tu es un assistant serviable. Réponds à la question de l'utilisateur en utilisant uniquement le nombre nécessaire. N'ajoute ni explications ni texte supplémentaire. Arrête la génération immédiatement après ta réponse à la question.
- es** Eres un asistente útil. Responde la pregunta del usuario usando solo el número necesario. No incluyas explicaciones ni texto adicional. Detén la generación inmediatamente después de responder la pregunta.
- pl** Jesteś użytecznym asystentem. Odpowiadaj na pytania użytkownika, używając jedynie liczby, która jest potrzebna. Nie dodawaj wyjaśnień ani dodatkowego tekstu. Przestań generować natychmiast po udzieleniu odpowiedzi.
- ru** Ты полезный помощник. Отвечай на вопросы пользователя, используя только необходимое число. Не добавляй пояснений или дополнительного текста. Останавливай генерацию сразу после ответа.
- ja** あなたは役に立つアシスタントです。ユーザーの質問には、回答に必要な数字だけを使って答えてください。説明や余分なテキストは含めないでください。質問に答えたらすぐに出力を終了してください。

P₆ - MKQA: number with unit questions (6 languages)

- en** You are a helpful assistant. Answer the user's question using only the required number and its unit. Do not include explanations or extra text. End your output immediately after your answer to the question.
- fr** Tu es un assistant serviable. Réponds à la question de l'utilisateur en utilisant uniquement le nombre demandé et son unité. N'ajoute ni explications ni texte supplémentaire. Arrête la génération immédiatement après ta réponse à la question.
- es** Eres un asistente útil. Responde la pregunta del usuario usando solo el número requerido y su unidad. No incluyas explicaciones ni texto adicional. Detén la generación inmediatamente después de responder la pregunta.
- pl** Jesteś użytecznym asystentem. Odpowiadaj na pytania użytkownika, używając tylko wymaganej liczby i jej jednostki. Nie dodawaj wyjaśnień ani dodatkowego tekstu. Przestań generować natychmiast po udzieleniu odpowiedzi.
- ru** Ты полезный помощник. Отвечай на вопросы пользователя, используя только необходимое число и его единицу измерения. Не добавляй пояснений или дополнительного текста. Останавливай генерацию сразу после ответа.
- ja** あなたは役に立つアシスタントです。ユーザーの質問には、必要な数値とその単位のみを使って答えてください。説明や余分なテキストは含めないでください。質問に答えたらすぐに出力を終了してください。

P₇ - Global MMLU (6 languages)

- en** You are a helpful assistant. Answer the user's question in one to ten words. If the answer is a number, respond with only that number. Do not include explanations or extra text. End your output immediately after your answer to the question.
- fr** Tu es un assistant serviable. Réponds à la question de l'utilisateur en un à dix mots. Si la réponse est un nombre, indique uniquement ce nombre. N'ajoute ni explications ni texte supplémentaire. Arrête la génération immédiatement après ta réponse à la question.
- es** Eres un asistente útil. Responde la pregunta del usuario en una a diez palabras. Si la respuesta es un número, responde solo con ese número. No incluyas explicaciones ni texto adicional. Detén la generación inmediatamente después de responder la pregunta.
- pl** Jesteś asystentem pomocniczym. Odpowiadaj na pytania użytkownika, używając od jednego do dziesięciu słów. Jeśli odpowiedź jest liczbą, podaj tylko tę liczbę. Nie dodawaj wyjaśnień ani dodatkowego tekstu. Przestań generować natychmiast po udzieleniu odpowiedzi.
- ru** Ты полезный помощник. Отвечай на вопросы пользователя, используя от одного до десяти слов. Если ответ число, укажи только это число. Не добавляй пояснений или дополнительного текста. Останавливай генерацию сразу после ответа.
- ja** あなたは役に立つアシスタントです。ユーザーの質問に1～10語で答えてください。答えが数字の場合は、その数字のみを答えてください。説明や余分なテキストは含めないでください。質問に答えたらすぐに応答を終了してください。

Figure 16: System prompts used for answer extraction, organized by dataset and by question type (P₁-P₅). All prompts cover six languages: English (en), French (fr), Spanish (es), Polish (pl), Russian (ru), and Japanese (ja).

P₈ - Correctness Assessment System Prompts (6 languages)

- en** You are an expert answer evaluator assessing whether a predicted answer conveys the key information as any of the correct answers, regardless of phrasing. Reply with only YES or NO.
- fr** Tu es un expert en évaluation des réponses, chargé de déterminer si une réponse proposée transmet les informations clés au même titre que n'importe quelle réponse correcte, indépendamment de sa formulation. Réponds par OUI ou NON uniquement.
- es** Usted es un evaluador experto de respuestas que debe determinar si una respuesta predicha transmite la misma información clave que alguna de las respuestas correctas, independientemente de la redacción. Responda solo con SÍ o NO.
- pl** Jesteś ekspertem w ocenie odpowiedzi i oceniasz, czy przewidywana odpowiedź przekazuje kluczowe informacje, jako którakolwiek z poprawnych odpowiedzi, niezależnie od sformułowania. Odpowiadaj tylko TAK lub NIE.
- ru** Вы являетесь экспертом по оценке ответов. Определите, передает ли предсказанный ответ ключевую информацию так же, как и любой из правильных ответов, независимо от формулировки. Отвечайте только «ДА» или «НЕТ».
- ja** あなたは専門の回答評価者です。予測された回答が、表現方法に関わらず、いずれかの正解と同等に重要な情報を伝えられているかを評価しています。「はい」または「いいえ」のみで回答してください。

Figure 17: System prompt for LLM judge used to assess answer correctness. All prompts cover 6 languages: English (en), French (fr), Spanish (es), Polish (pl), Russian (ru), and Japanese (ja).

P₉ - P(True) Prompts (6 languages)

- en** Question: {question}
Possible answer: {answer}
Is the possible answer:
(A) True
(B) False
You must respond with only (A) or (B). The possible answer is:
- fr** Question : {question}
Réponse possible : {answer}
La réponse possible est-elle :
(A) Vrai
(B) Faux
Vous devez répondre uniquement par (A) ou (B). La réponse possible est :"
- es** Pregunta: {question}
Posible respuesta: {answer}
La posible respuesta es:
(A) Verdadero
(B) Falso
Debes responder únicamente con (A) o (B). La posible respuesta es:
- pl** Pytanie: {question}
Możliwa odpowiedź: {answer}
Czy możliwa odpowiedź to:
(A) Prawda
(B) Fałsz
Musisz odpowiedzieć wyłącznie (A) lub (B). Możliwa odpowiedź to:
- ru** Вопрос: {question}
Возможный ответ: {answer}
Верен ли возможный ответ:
(A) Верно
(B) Неверно
Необходимо ответить только (A) или (B). Возможный ответ верен::
- ja** 質問: {question}
考えられる回答: {answer}
考えられる回答は:
(A) 真
(B) 偽
(A)または(B)のみで答えてください。考えられる回答は:

Figure 18: Prompts used for P(True) (Kadavath et al., 2022). All prompts cover 6 languages: English (en), French (fr), Spanish (es), Polish (pl), Russian (ru), and Japanese (ja).

P₁₀ - Verbalized Confidence Prompts (6 languages)

- en** Considering the question and your provided answer, estimate the probability the answer is correct as a percentage from 0 to 100, where 0 = certainly incorrect and 100 = absolutely correct. Return only Confidence: <0-100>.
- fr** Compte tenu de la question et de votre réponse, estimez la probabilité que la réponse soit correcte, en pourcentage, entre 0 et 100, où 0 signifie « certainement incorrect » et 100 « absolument correct ». Indiquez uniquement le niveau de confiance : <0-100>.
- es** Considerando la pregunta y la respuesta proporcionada, estime la probabilidad de que la respuesta sea correcta en forma de porcentaje de 0 a 100, donde 0 = definitivamente incorrecta y 100 = absolutamente correcta. Devuelva solo el nivel de confianza: <0-100>.
- pl** Biorąc pod uwagę pytanie i udzieloną odpowiedź, oszacuj prawdopodobieństwo, że odpowiedź jest poprawna, w procentach od 0 do 100, gdzie 0 oznacza odpowiedź zdecydowanie błędną, a 100 – odpowiedź całkowicie poprawną. Zwróć tylko poziom pewności: <0-100>.
- ru** Учитывая вопрос и предоставленный вами ответ, оцените вероятность правильности ответа в процентах от 0 до 100, где 0 = заведомо неверный ответ, а 100 = абсолютно верный ответ. Верните только Уверенность: <0-100>.
- ja** 質問とあなたの回答を考慮し、回答が正しい確率を0100の範囲で推定してください。0は確実に間違っている、100は絶対に正しいとします。信頼度<0-100>のみを返してください。

Figure 19: Prompts to ask for verbalized confidence. All prompts cover 6 languages: English (en), French (fr), Spanish (es), Polish (pl), Russian (ru), and Japanese (ja).